

Status of the Gunter & Scott 1990 Development in Lean

main at 5f2ac51. Build 1372 jobs, 0 errors, 3 sorry warnings, 0 other warnings. 124 modules, 46,442 lines, 2,122 theorems.

1. Extractions — what the paper contains

#	Kind	Count
1	Definitions (unnumbered)	11
2	Numbered theorems	16
3	Numbered lemmas	13
4	Numbered propositions	1
5	— numbered results, total	30
6	— conjuncts those 30 decompose into	93
7	Prose statements	146
8	Examples	1
9	Figures	4

2. Represented in Lean

#	Kind	Represented	Of	Not stated
1	Definitions	11	11	0
2	Theorems	16	16	0
3	Lemmas	13	13	0
4	Propositions	1	1	0
5	Prose statements	86	146	60
6	Examples	1	1	0

3. Corrected — could not be transcribed as printed

#	Kind	Printing	Reading	Semantic	Corrected	— proven	— unproven
1	Definitions	1	0	1	2	2	0
2	Theorems	1	1	2	4	2	2
3	Lemmas	1	2	0	3	2	1
4	Propositions	0	0	0	0	0	0
5	Examples	0	0	0	0	0	0
6	— total	3	3	3	9	6	3
7	Prose statements	—	—	—	—	—	not classified

Printing error — the page is garbled; the mathematics is right

#	Where	What	Corrected to	Proven
1	Lemma 9	the PDF drops every $\otimes$ and every $\perp$, so which operators the laws range over is unreadable	lem9_1...lem9_6; items 3 and 5 also carry kernel-checked negations of the printed forms	yes
2	Theorem 14	the list of characterizations is garbled	thm14, both directions	yes
3	§7.4 worked example	reverses the paper's own definition — printed $b \vdash a$ where both papers give $a \vdash b$	read in the paper's direction; the paper's counts 1, 2, 5, 20 select it over the rival	yes

Reading error — ours; the paper is right

#	Where	What	Corrected to	Proven
1	Theorem 29, sentence 2	dropped “domain” from “any bifinite domain”, making it false	Thm29SecondAtDomains	no
2	Lemma 28	quantified over all U; the paper states it over its own U	PRep.Lemma28AtU, nine conjuncts, no hypotheses	yes
3	Lemma 30	same shape — a universal closure the paper does not claim	Lemma30AtV, arity 1	no

Semantic error — the mathematics is wrong as stated

#	Where	Whose	What	Corrected to	Proven
1	Theorem 26	paper	false for any signature with two or more 0-ary slots, including the paper’s own $(2, 0, 0, 0, 0, 0)$	thm26 with $hs : \forall i, 0 < s\ i$	yes
2	Theorem 7	ours	guarded on compactness where the sentence is about boundedness; on the paper’s index the join always exists, so it tests neither	StepFunctionsDecidable over IsStepEnumeration, existentially quantified	no
3	§7.4 order relation	paper	not reflexive — $b = (\perp, \emptyset)$ fails $b \vdash b$; it is the strict part	reflexive closure; Colimit.V built on it, isoPlus : $V \approx_0 \text{Plus } V$	yes

4. Proof state

#	Kind	Proven	Unproven	Of
1	Theorems	14	2	16
2	Lemmas	12	1	13
3	Propositions	1	0	1
4	— numbered results	27	3	30
5	Prose statements	70	16	86 stated
6	Examples	1	0	1

#	Unproven result	State
1	Theorem 7	sentences 1 and 2 proven; the recursive forms are not
2	Theorem 29	sentence 1 proven; sentence 2 is not
3	Lemma 30	stated at V, arity 1 — one named input left

All three reduce to **three** statements, each carried by Gunter & Scott in 1990. Each is a Lean theorem ending in sorry, so the build reports it (r0052):

#	The unproven statement	Lean theorem	Blocks
1	RecursiveNormal for $K(D \rightarrow E)$ — decide whether two basis elements are bounded, and compute their join’s index	R49.Agent3.scottHomCRecursive_unproven	Theorem 7, arrow
2	the same over $K(D \multimap E)$ — no proved reduction carries recursiveness across $K(D \multimap E) \hookrightarrow K(D \rightarrow E)$	R49.Agent3.strictHomCRecursive_unproven	Theorem 7, strict
3	Theorem29Normal at infinite $K(E)$ — realize the type over the tower’s image, not the η -image; finite bases are proven	LemThirty.theorem29Normal_unproven	Theorem 29, Lemma 30

5. Proof work state

#	Kind	Left with a sorry
1	Definitions	0
2	Theorems	3
3	Lemmas	0
4	Propositions	0
5	Prose statements	0
6	Examples	0

sorry 3, axiom 0, since r0052; sorry was 0 from r0042 to r0051. Every unproved result is a theorem ... := sorry, so the unproven count **is** the sorry count: the three of table 4's second table above. Exactly three package constants depend on sorryAx — those three theorems — and no other declaration does, because none applies them: every consumer still takes its claim as an explicit hypothesis.

6. Size

#	Quantity	Count
1	Files	124
2	Modules	124
3	Lines	46,442
4	Theorems	2,122
5	defs	829
6	— of those, Prop-valued	120
7	Structures and classes	25
8	Constants in the environment	4,393
9	Build jobs	1,372

7. Mathlib theories referenced

34 distinct Mathlib modules, across 8 top-level theories.

#	Theory	Modules
1	Mathlib.Data	12
2	Mathlib.Order	11
3	Mathlib.SetTheory	3
4	Mathlib.Algebra	3
5	Mathlib.Logic	2
6	Mathlib.Computability	1
7	Mathlib.Combinatorics	1
8	Mathlib.Tactic	1

Mathlib.Order.CompletePartialOrder and Mathlib.Order.ScottContinuity are the two the development is built on; Mathlib.Computability.RE is the only one §3.2 needs.

Reproducing

```
scripts/counts.sh           # modules, lines, theorems, sorry
scripts/numbered-status.sh  # declarations per numbered result
scripts/a6-env-scan.sh <out> # then a6-summarize.py
scripts/a8-claim-check.sh   # prose staleness against the environment
scripts/compile.sh -r <round> # build
scripts/mathlib-imports.sh   # tables 6 and 7
```

Tables 1 and 9 come from pdftotext -layout on the paper. The text layer drops the fi ligature — Definition reads De nition, bifinite reads bi nite — so a grep for Definition returns 0 and is wrong.